

UNIT 03

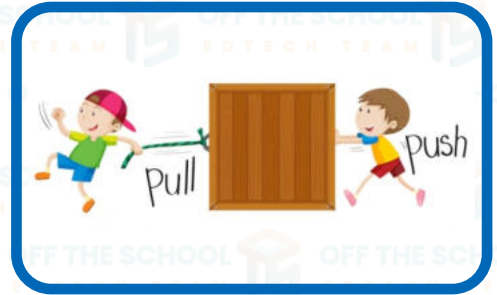
DYNAMICS



DYNAMICS

“Dynamics is the study of how objects move and the forces that affect their motion.”

It includes the study of how objects move and change over time, as well as the laws and principles that govern that motion. Dynamics helps to explain and predict the motion of a wide range of physical systems, including everything from subatomic particles to galaxies.



FORCE

Force is the agent that changes the state of rest or uniform motion of a body.

UNIT

Its SI unit is Newton (**N**)



PROPERTIES

- ▶ Force is required to change the position.
- ▶ Force can act as a pull or push agent.
- ▶ Force produces acceleration.
- ▶ It can produce distortion.

ONE NEWTON (1 N)

One Newton (**1 N**) is the amount of force that can produce 1 m/s^2 acceleration in 1 Kg mass.

CLASSIFICATION OF FORCE

Forces can be classified into several types, such as contact forces, field forces, and action and reaction forces. Friction, gravity, and electrostatic forces etc.

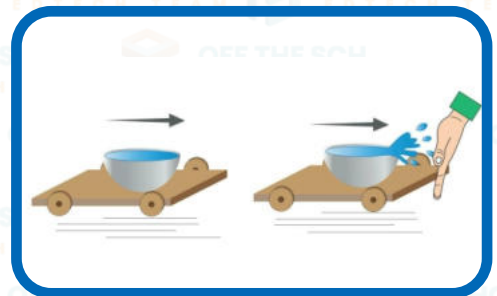
NEWTON'S FIRST LAW OF MOTION

Newton's first law of motion, also known as the law of inertia, states that:

“An object at rest will remain at rest, and an object in motion will remain in motion with a constant velocity, unless acted upon by an external force.”

You have often observed the table placed in your classroom.

It always remains at the same place until you apply some force to move it. Contrary to the above example, a ball rolling on the ground however stops after some time because friction of ground and air resistance exert force on it and change its state of motion or direction of motion.



LAW OF INERTIA

“Inertia is the tendency of an object to remain at rest or continue moving at a constant speed in a straight line unless acted upon by an external force.”

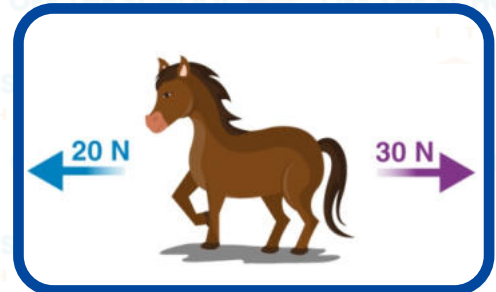
FOR EXAMPLE

An example of inertia is a passenger in a moving car feeling pushed back into their seat when the car suddenly stops, as the passenger continues to move forward due to their inertia.

NEWTON'S FIRST LAW OF MOTION

Newton's second law of motion states that:

"The acceleration of an object is directly proportional to the net force applied to it and inversely proportional to its mass."

**Mathematically**

$$a \propto F \quad \text{(i)}$$

$$a \propto \frac{1}{m} \quad \text{(ii)}$$

After combining both equations

$$a \propto \frac{F}{m}$$

$$ma \propto F$$

$$F = kma$$

Where k is the constant of value 1 then,

$$F = (1) ma$$

$$F = ma$$

THIRD LAW OF MOTION

Newton's third law of motion states that:

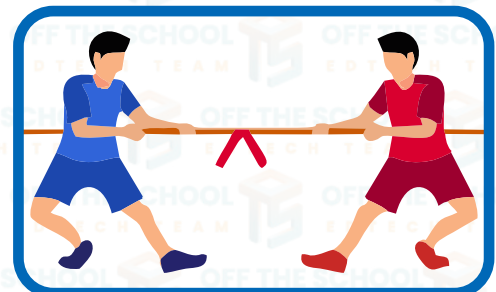
"To every action, there is an equal and opposite reaction".

Action = - Reaction

The action and reaction are forces that occur together as a pair. They are always equal in quantity but opposite in direction.

For example

- ▶ Standing on the ground the gravity pulls you down against the ground, the ground pushes up against your feet.
- ▶ When a rocket ignites its fuel behind it, the expanding exhaust gas pushes on the rocket causing it to accelerate.



MASS AND WEIGHT

MASS

Mass is the actual amount of material contained in a body. It is measured in kg. Mass is independent of everything. Mass is an intrinsic property of the body and remains the same wherever the body might be.

WEIGHT

Weight is the force exerted by the gravity on that object. Weight is measured in Newton (N). Weight is different on the earth, moon, and other places due to difference of gravitational pull. Weight is the measure of how strongly gravity pulls on that matter.

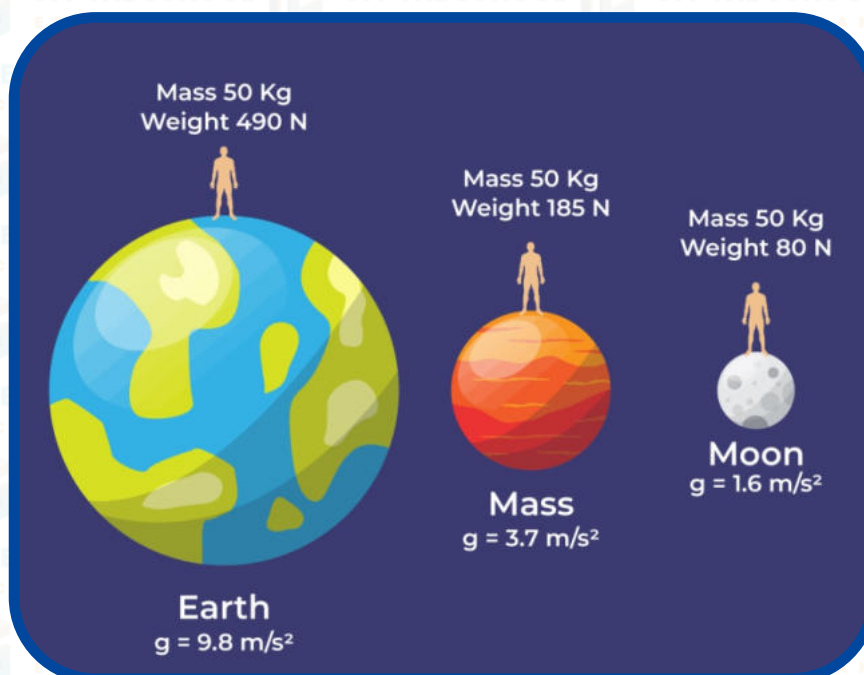
DIFFERENCE BETWEEN MASS AND WEIGHT

MASS

- ▶ Mass is the quantity of matter in a body regardless of its volume or of any forces acting on it.
- ▶ It is measured in kg.
- ▶ It is a scalar quantity.
- ▶ Mass is an intrinsic property of the body and remains the same wherever the body might be.

WEIGHT

- ▶ Weight is a measurement of the gravitational force acting on an object.
- ▶ Weight is measured in Newton (N).
- ▶ It is a vector quantity.
- ▶ Weight is different on the earth, moon, and other places due to difference of gravitational pull.



MOMENTUM

The momentum of an object is equal to the mass multiplied by the velocity of the object.

Mathematically

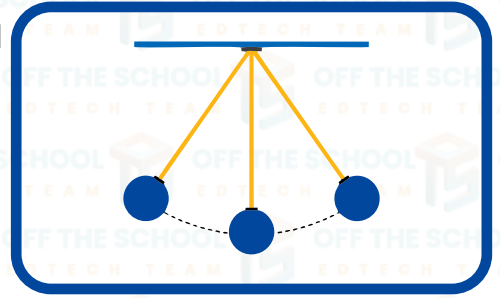
Momentum = mass x velocity

$$P = m \times v$$

The momentum depends upon the quantity of mass and velocity of the object. Greater the mass of the body, greater would be the momentum, Greater the velocity of the body of the body, greater would be the momentum.

For example

Momentum of a moving train, which has a large amount of momentum due to its larger mass and velocity, and it is difficult to stop or change its direction.



UNIT

The SI unit of momentum is Ns (Newton second)

MOMENTUM IN TERM OF FORCE

The change in momentum is equal to the force multiplied by the time interval for which it was applied.

EXPLANATION

Consider a body of mass (m), moving with initial velocity (v_i), A force (F) acts on the body to produce acceleration (a) therefore the final velocity after time (t) will become (v_f).

If $P = mV$ and m is constant, then the change in velocity changes the momentum of the body.

Initial momentum = Final momentum

$$P_i = P_f$$

$$mV_i = mV_f$$

$$\Delta P = P_f - P_i$$

$$P_f - P_i = mV_f - mV_i$$

$$P_f - P_i = m(V_f - V_i)$$

Divide both sides with t

$$\frac{P_f - P_i}{t} = m \frac{V_f - V_i}{t}$$

As the rate of change of velocity is acceleration

Therefore,

$$\frac{V_f - V_i}{t} = a$$

$$\frac{P_f - P_i}{t} = ma$$

According to the Newton's second law of motion $F = ma$

$$F = \frac{P_f - P_i}{t}$$

$$F = \frac{\Delta P}{t}$$

LAW OF CONSERVATION OF MOMENTUM

MOMENTUM

"The total momentum of an isolated system always remains constant"

EXPLANATION

Conservation of momentum states that the total momentum of a closed system remains constant, unless acted upon by an external force. This means that if two or more objects interact, the total momentum before the interaction will be equal to the total momentum after the interaction. This principle is useful in many fields, including physics, engineering, and sports, to understand and predict motion and changes in motion. For simplicity consider a system of two billiard balls of masses m^1 and m^2 moving in straight line with velocities U^1 and U^2 respectively where U^2 is greater than U^1 . Total momentum before collision = $m^1 U^1 + m^2 U^2$ After collision the velocities become V^1 and V^2 respectively, Total momentum after collision = $m^1 V^1 + m^2 V^2$ According to law of conservation of momentum: Total momentum of the system before collision = Total momentum of the after collision $m^1 U^1 + m^2 U^2 = m^1 V^1 + m^2 V^2$

UNIFORM CIRCULAR MOTION

Uniform circular motion refers to the motion of an object moving in a circular path at a constant speed. In this type of motion, the speed of the object is constant and the direction of motion is constantly changing, as the object moves along the circumference of the circle.

For example

A common example of uniform circular motion is a satellite orbiting around the Earth. The satellite is moving in a circular path around the Earth at a constant speed, but its direction of motion is constantly changing due to the Earth's gravitational pull.

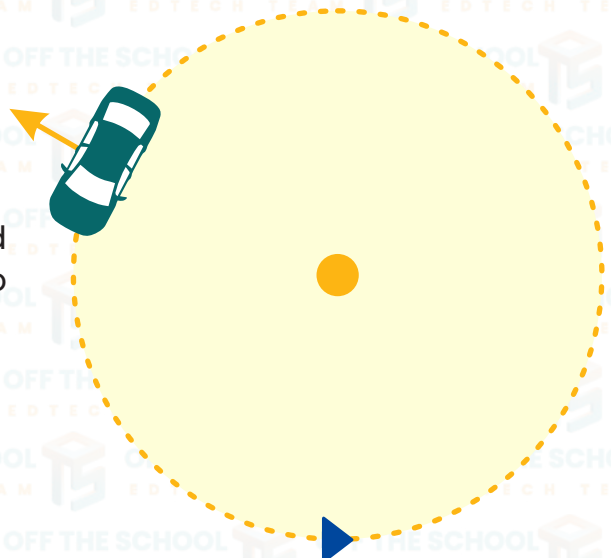


CENTRIPETAL FORCE

The force required to move a body along a circular path is called Centripetal force. It is denoted by F_c . The centripetal force is always directed towards the centre of the circle. It depends on three factors: The velocity of the object (v). The object distance from the centre " r ". The mass of the object " m ".

$$F_c = \frac{mv^2}{r}$$

The velocity of the object is constant and perpendicular to a line running from the object to the centre of the circle.



For example:

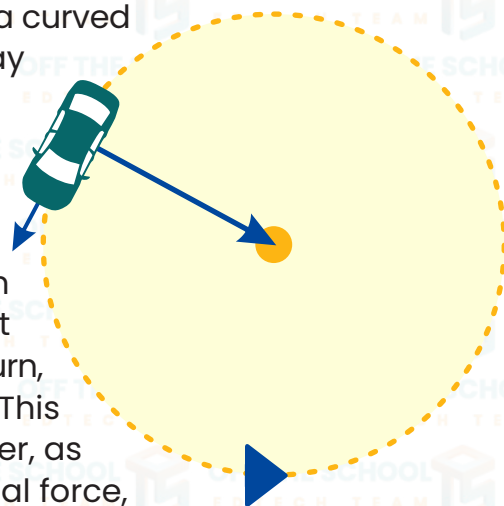
A ball tied to a string and being swung around in a circle experiences a force directed towards the centre of the circle, which is the tension in the string. This force is the centripetal force, responsible for keeping the ball in uniform circular motion. If the string is suddenly released, the ball would no longer experience the centripetal force, resulting in the ball moving off in a straight line tangent to the circle. Thus, the centripetal force is necessary to maintain an object in uniform circular motion and to change its direction.

CENTRIFUGAL FORCE

A force that acts outward on a body which moves along a curved path is called centrifugal force. It is always directed away from the centre of the curvature. The magnitude of centrifugal force is equal but opposite in direction to centripetal force.

For example:

A simple example of centrifugal force in action can be seen when driving a car around a sharp turn. As the car turns, it experiences a centripetal force towards the centre of the turn, provided by the friction between the tires and the road. This force allows the car to maintain its circular path. However, as the car turns, an equal and opposite force, the centrifugal force, acts on the car, pushing it outwards and away from the centre of the turn. If the turn is sharp enough or the speed is high enough, the centrifugal force can become strong enough to overcome the frictional force, causing the car to skid out of control.

**APPLICATIONS OF CENTRIFUGE**

Centrifuge appliances are used to separate heavier particles from lighter particles in liquids e.g. Sugar crystals are separated from molasses. Blood analysis is carried out through a centrifuge process in the laboratory. Cream separator is used to separate the cream from skimmed milk. An ultracentrifuge is used for separating small particles from large molecules. A centrifuge is used for separation of isotopes

ROAD BANKING

The outer edge or bank of the road is raised to a certain height at the curved part of roads. This provides the centripetal force against the tyres of the vehicle hence prevents skidding.

CREAM SEPARATOR

The milk plants in the country are using high speed spinners to separate cream from milk. The skimmed milk is heavier whereas the cream is lighter. When the milk is spun at high speed the heavy particles are pushed towards the walls of the spinner. These particles push the lighter particles of cream to the centre where from it is collected through a tube.

DRYER

Now a day's built-in dryer is available in most washing machines. It spins the wet clothes hence the water droplets are thrown away from the perforated walls of the dryer and clothes get dry instantly.

FRICTION

The force that resists relative motion between two surfaces is called friction.

$$F_s \propto R \quad \text{(i)}$$

The ratio between limiting force and the normal reaction force R is constant that is represented by coefficient of friction, thus

$$F_s = \mu R \quad \text{(ii)}$$

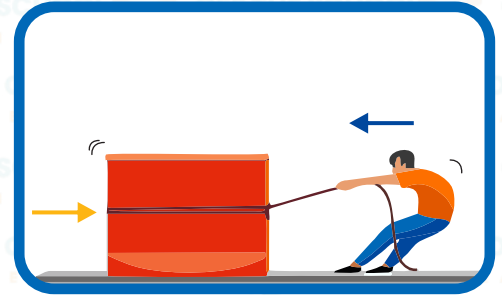
OR

$$\mu = \frac{F_s}{R}$$

When a body is placed on a surface its weight w acts downward then according to Newton's third law of motion $R = W$, here $W = mg$ by putting this value in equation (ii).

$$F_s = \mu mg$$

The coefficient of friction has different values for different surfaces.



TYPES OF FRICTION

STATIC FRICTION

The force that acts on a stationary object and opposes its ability to start moving is known as static friction. The maximum value of static friction is referred to as "limiting friction."

KINETIC FRICTION

The force that opposes the motion of a moving object is known as friction. It is noteworthy that, in nearly all circumstances, the force of static friction exceeds that of kinetic friction.

SLIDING FRICTION

When one body slides over another body, the friction between the two surfaces is referred to as sliding friction.

ROLLING FRICTION

When a body moves on wheels, the friction is called rolling friction, which is much less than sliding friction.

ADVANTAGES OF FRICTION

👁 **Enables Movement:** Friction allows for traction, which is essential for movement, whether it's walking, running, or driving a car.

- **Prevents Slippage:** Friction helps to prevent slippage between two surfaces, which is important in many industrial and everyday applications, such as machinery and tools.
 - **Aids in Braking:** Friction between the brake pads and the wheel helps to slow down and stop a vehicle.
 - **Facilitates Grip:** Friction allows us to grip objects and hold onto them, which is important for tasks such as writing, holding a baseball bat, and performing manual labour
 - **Essential for Heat Generation:** Friction is used to generate heat and energy in many industrial processes, such as grinding and drilling.
- In summary, friction plays an important role in many everyday activities and industrial processes

DISADVANTAGES OF FRICTION

- **Wear and Tear:** Friction causes wear and tear on machinery and equipment, which can lead to breakdowns and the need for repairs.
- **Energy Loss:** Friction consumes energy, reducing the efficiency of machines and processes.
- **Heat Generation:** Friction can generate heat, which can be dangerous in certain industrial settings, such as in the production of machinery and equipment.
- **Difficulty in Movement:** Friction can make it difficult to move objects, such as a car, which can be especially challenging in slippery conditions like snow and ice.
- **Hazardous Work Environment:** Friction can create hazardous work environments, such as in mining and construction, where workers may be exposed to heat and sparks generated by friction.

Overall, friction can be both helpful and harmful, and it's important to understand its effects to make the most of its benefits and minimize its drawbacks.

WAYS TO REDUCE FRICTION

There are several ways to reduce friction:

- **Lubrication:** By applying a lubricant such as oil or grease between two objects, friction can be reduced.
- **Surface Smoothing:** Smoothing the surface of an object can reduce friction because it creates fewer points of contact between the objects.
- **Increasing the Surface Area:** Increasing the surface area of the objects in contact can reduce friction. For example, using a larger wheel reduces friction compared to a smaller wheel because it has a larger surface area in contact with the ground.
- **Change in Materials:** Friction can be reduced by changing the materials in contact. For example, using a low friction material such as Teflon can reduce friction compared to a high friction material such as rubber.
- **Using Ball Bearings:** Ball bearings reduce friction by allowing an object to rotate smoothly on a shaft.



NOTE

These are the general ways to reduce friction, the actual methods may vary based on the specific scenario and application.